LAMP: Lift Image-Editing as General 3D Priors for Open-world Manipulation

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Figure 1. We propose **LAMP**, which lifts image editing as general 3D priors, enabling open-world manipulation of diverse tasks from monocular RGB-D observations and promptable instructions.

Abstract

Human-like generalization in open-world remains a fundamental challenge for robotic manipulation. Existing learning-based methods, including reinforcement learning, imitation learning, and vision-language-action-models (VLAs), often struggle with novel tasks and unseen environments. Another promising direction is to explore generalizable representations that capture fine-grained spatial and geometric relations for open-world manipulation. While large-language-model (LLMs) and vision-language-model (VLMs) provide strong semantic reasoning based on language or annotated 2D representations, their limited 3D awareness restricts their applicability to fine-grained manipulation. To address this, we propose LAMP, which lifts image-editing as 3D priors to extract inter-object 3D transformations as continuous, geometry-aware representations. Our key insight is that image-editing inherently encodes rich 2D spatial cues, and lifting these implicit cues into 3D transformations provides fine-grained and accurate guidance for open-world manipulation. Extensive experiments demonstrate that LAMP delivers precise 3D transformations and achieves strong zero-shot generalization in open-world manipulation.

1. Introduction

Achieving human-like generalization in open-world robotic manipulation remains one of the ultimate goals for embodied intelligence. The challenge stems from the wide variety of task structures, levels of complexity, and temporal horizons. Traditional methods typically rely on task-specific modeling of robot states [8, 49, 57], which limits their generalizability. Recent learning-based approaches like reinforcement learning (RL) [2, 24, 25, 27, 37, 48, 55, 80, 82], imitation learning (IL) [15, 32, 56, 61], and VLAs [5, 10, 11, 35, 39, 51, 66, 89] adopt a data-driven paradigm by

training networks on various robot data. But they struggle to handle novel tasks and environments that are entirely different, falling short in open-world manipulation. To reach the goal, another strategy is to explore a generalizable representation for robotic manipulation in open worlds.

One promising direction for open-world manipulation is to leverage the spatial reasoning ability of LLMs [1, 3, 70, 71] and VLMs [4, 23, 44, 75]. Some works [30, 41] leverage the code-generation capability of LLMs to represent manipulation as executable code segments from language instructions. This representation effectively converts simple and concrete spatial expressions (e.g., "move up", "go left", "1 meter away") into actionable code primitives, yet lacks perception of the actual scene and geometry due to the absence of visual grounding. Other methods [20, 29, 31, 50, 52] instead represent manipulation as geometric relations (e.g. distance, parallelism, or perpendicularity) between annotated entities (e.g., keypoints or vectors) on 2D observations. While effective for simple spatial reasoning, these explicit 2D annotations are fragile under noisy depth and viewpoint changes. Despite their difference, both LLM- and VLM-based previous approaches ultimately rely on language-described explicit constraints that are inherently sparse and ambiguous in 3D space. They struggle to express fine-grained geometric relations, such as relative rotations, contact geometry or precise alignment between interacting objects, which are essential for precise manipulation like assembly [13]. The core limitation stems from the discrete and symbolic nature of language, which makes it hard to capture continuous 3D spatial interactions.

To address this challenge, we seek a representation that captures continuous and geometry-aware spatial relations beyond discrete linguistic constraints and remains robust to viewpoint variations. Inspired by assembly tasks, we adopt inter-object 3D transformations as a physically grounded representation for manipulation. Such transformations naturally encode relative motion, contact geometry, and alignment between objects in 3D space. However, obtaining accurate 3D priors remains nontrivial. Video- [6, 7, 53] and 4D-generative models [88] provide a potential path to extract such priors, but currently they still suffer from severe visual inconsistency and incorrect functional understanding, while being computationally expensive. We instead observe that image-editing models implicitly encode rich spatial priors in the 2D visual domain: how an object should move, rotate, or interact within a scene. Moreover, due to their paired image supervision and object-consistent editing behavior, these models maintain strong subject consistency across edits. This motivates our central question: *Can we extract 3D priors for manipulation from image editing?*

We introduce **LAMP** (**L**ift Im**A**ge-Editing as General 3D Priors for Open-World **M**ani**P**ulation). It lifts spatial clues in edited images into 3D inter-object transformations. Specifically, given a task instruction, we first perform image editing on the current observation to obtain an edited state. Using the current depth map from a RGB-D camera and single-view reconstruction [74], we lift the current and edited states into their 3D coordinate frames, and compute their inter-object transformation by aligning the active and passive manipulation objects of the current frame to the edited frame. This dense 3D transformation acts as a continuous geometric prior, encoding both spatial alignment and interaction intent. We enhance robustness to depth noise with 2D-3D fused hierarchical point-cloud filtering, which retains only reliable partial geometry under viewpoint variation. We further handle the potential inter-object scale inconsistency introduced by image-editing [14] (e.g., object size change between the current and edited states) via scale alignment. Our main contributions are as follows:

- We propose LAMP, which lifts image-editing into 3D general priors for manipulation and extracts precise inter-object 3D transformations from single-view RGB-D observations, enabling efficient open-world manipulation.
- We provide an in-depth analysis of current VLM/LLM-based open world manipulation methods and demonstrate the superior generalization and robustness of our image-editing-lifted 3D priors in open world settings.
- Through extensive experiments, we demonstrate our method's strong zero-shot generalization across a diverse variety of real-world manipulation tasks.

2. Related Works

General Representations for Robotic Manipulation. General representations are the key to achieving strong generalization in open-world manipulation. Traditional end-to-end policies learn neural descriptors as object-centric representations for downstream control [16, 26, 63, 64, 69, 81], enabling in-category generalization. For open world manipulation, recent efforts instead annotate 2D observation with keypoints [20, 31, 50], projected vectors [29], and 3D poses [52] to provide structured visual inputs for VLMs and LLMs. While interpretable for LLMs or VLMs, these 2D explicit representations are often brittle under occlusion, viewpoints shifts, and depth noise, leading to unstable grounding across diverse scenes. Another direction learns object poses or bounding boxes as an intermediate representation [45, 72], but pose estimation can struggle to generalize across unseen objects. An alternative line of work focuses on learning inter-object transformations as generalizable representations for assembly tasks [13, 47, 54, 67, 77, 79, 86], but such methods depend on complete 3D inputs and task-specific training. In this paper, we lift 2D image-editing into 3D transformation, providing a spatially grounded representation that remains stable under real-world noise and partial observations.

Foundation Models for Manipulation. Foundation mod-

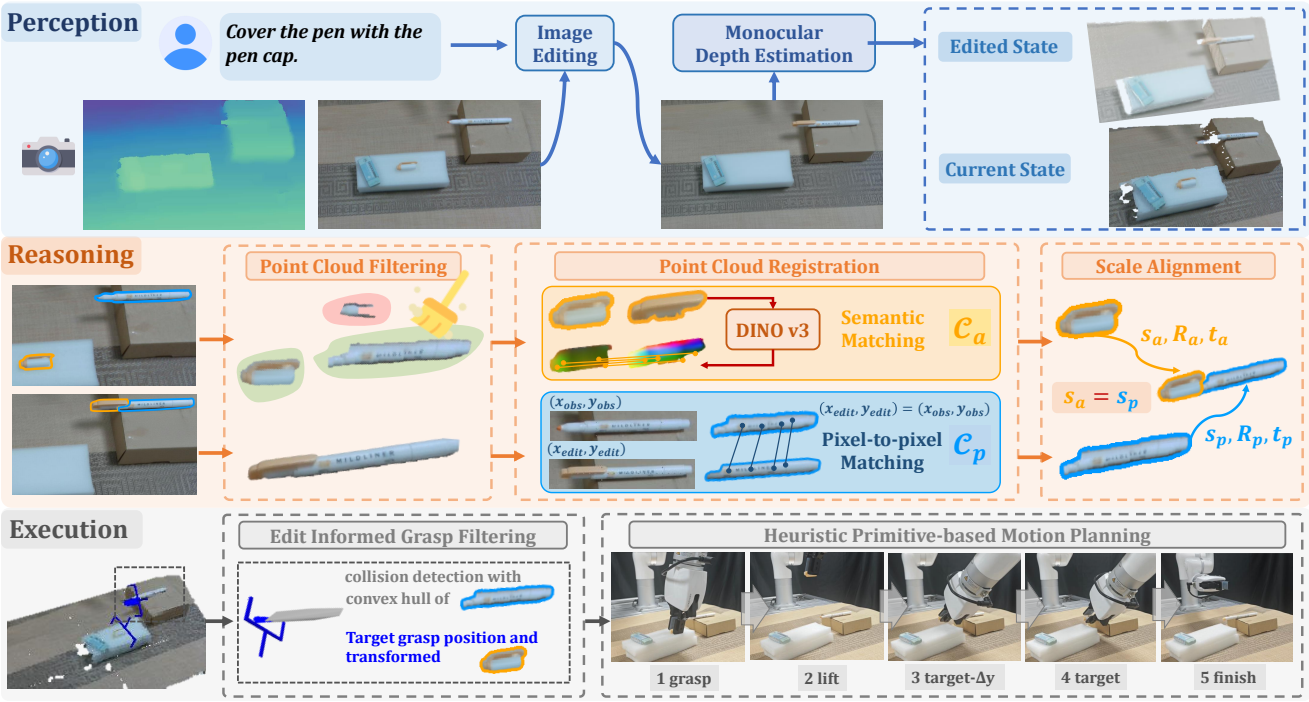

Figure 2. **Overview.** Given the RGB-D observation and a language instruction, the Image-editing generates an edited state, which is used for registration to extract the inter-object transformation in reasoning stage. This transformation is converted into target pose for execution.

els leverage large-scale vision-language priors for embodied reasoning [21, 34]. VLM- and LLM-based methods [20, 29–31, 40, 50, 52] use spatial cues (e.g., 3D action maps [30], relational keypoints [31], or interaction vectors from images [29]) to ground manipulation behavior but remain limited especially for fine-grained control, due to the sparsity of language constraints and ambiguity of 2D grounding in 3D scenes, To overcome this, VLA [10, 11, 38, 85, 87, 89] and video-based approaches [6, 7, 18, 42, 53, 84, 88] learn action-level supervision from large robot data or human videos. In parallel to these paradigms, SuSIE [9] further uses image-editing diffusion models (e.g., Instruct-Pix2Pix [12]) to generate subgoal images that visually guide goal-conditioned policy. Unlike SuSIE's 2D formulation, we grounds visual editing in inter-object 3D transformations for general manipulation.

3. Method

At the core of our approach lies a simple question: can image editing provide stronger spatial priors for manipulation? Edited images implicitly specify how objects should move and relate spatially. This insight motivates us to formulate manipulation as predicting the inter-object 3D transformations (Sec. 3.1) and design a perception–reasoning–execution framework that converts visual edits into executable trajectories.

Overview. An overview of our pipeline is illustrated in Fig. 2. In the perception stage, we extract 3D spatial priors from the edited image to ground the high-level intent (Sec. 3.2). In the reasoning stage, we propose a noise-robust cross-state point cloud registration for real-world settings, enabling reliable estimation of the 3D inter-object transformation via the edited state (Sec. 3.3). Finally in the execution stage, the estimated transformation is converted into the target pose to optimize the end-effector trajectory (Sec. 3.4).

3.1. Task Formulation from an Editing Perspective

We formulate robotic manipulation as predicting the relative transformation of objects via visual editing. Given an initial RGB-D observation (I_{obs}, D) and a free-form language instruction $\mathcal{L}$, our goal is to generate a 6-DoF end-effector trajectory τ that executes the intended manipulation. The instruction $\mathcal{L}$ specifies a subtask-level manipulation rather than a high-level long-horizon command. Complex tasks can be decomposed into subtasks using a high-level planner [28, 65]. Specifically, $\mathcal{L}$ describes either a single target object $\mathcal{O}_a$ to be manipulated (e.g., "open the red drawer"), or an interaction between an active and a passive object $(\mathcal{O}_a, \mathcal{O}_p)$ (e.g., "cover the teapot with the lid"). Leveraging the inherent spatial reasoning embedded in image editing, we formulate each manipulation as predicting a target relative transformation $\mathbf{T}_a \in \text{SE}(3)$ of the active object $\mathcal{O}_a$, mapping it from the observed state to the edited state.

3.2. Spatial Prior Extraction from Editing

Given the current RGB observation $I_{\text{obs}} \in \mathbb{R}^{H\times W\times 3}$ and a task description $\mathcal{L}$, we generate an edited image $I_{\text{edit}} \in$

$\mathbb{R}^{H\times W\times 3}$ conditioned on $\mathcal{L}$ using modern image-editing models [17, 78] to depict the target post-manipulation state of the active object $\mathcal{O}_a$ visually. To recover its geometry, we lift I_{edit} into a pixel-aligned point cloud $\mathcal{P}_{\text{edit}} \in \mathbb{R}^{(H\times W)\times 3}$ using a monocular depth estimator (e.g., VGGT [74]). However, resolution mismatch between I_{edit} and the depth estimator may cause spatial detail loss if directly processed. To mitigate this, we extract binary masks $\mathcal{M}^a_{\text{edit}}$ and $\mathcal{M}^p_{\text{edit}}$ of $\mathcal{O}_a$ and $\mathcal{O}_p$ from I_{edit}, using LLMDet [22] for language-grounded localization and SAM [36] for pixel-level refinement. For single-object instructions (e.g., "open the red drawer"), the passive object $\mathcal{O}_p$ denotes its functionally coupled static surroundings (e.g., the drawer housing). We then crop the tight bounding box enclosing $\mathcal{M}^a_{\text{edit}}$ and $\mathcal{M}^p_{\text{edit}}$, and resize or pad it by the original I_{edit} to match the estimator's input resolution. For resized images, the predicted depth is upsampled back to the cropped RGB resolution, ensuring one-to-one pixel correspondence. This preserves spatial detail and yields accurate 3D grounding of manipulated regions in $\mathcal{M}^a_{\text{edit}} \cup \mathcal{M}^p_{\text{edit}}$ in I_{edit}.

3.3. Cross-state Point Cloud Registration

To estimate the 6-DoF transformation $\mathbf{T}_a$ of the active object $\mathcal{O}_a$, we register current and edited point clouds. While registration is well-studied in reconstruction [83], applying across edited states is challenging: observations are noisy and incomplete (Fig. 3(b)), and interacting objects $(\mathcal{O}_a, \mathcal{O}_p)$ may move, deform or occlude each other (Fig. 3(a)). To handle these issues, we propose a cross-state registration pipeline that sequentially filters unreliable points, performs object-centric alignment, and applies unified scale correction to maintain consistent spatial reasoning.

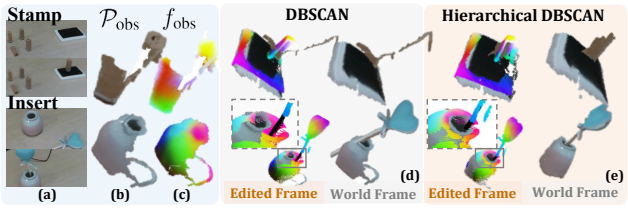

Figure 3. Illustration of the 2D-3D hierarchical point-cloud filtering. Outliers are spatially near inliers (b), but distant from inliers with similar features (c). Density-based clustering fails (d), while hierarchical filtering successfully remove them (e).

Point Cloud Filtering. RGB-D sensors often produce floating edge points due to depth discontinuities and sensor blur (Fig. 3(b)). Such artifacts degrade the accuracy of registration, especially for scale-sensitive manipulation. Classic density-based filters (e.g., DBSCAN [60]) may fail to remove them, because these artifacts remain locally dense and close to valid regions (Fig. 3(b)). Even depth-refinement methods [43] still output spatially coherent flying points once lifted into 3D. We observe that, while these flying-edge points are spatially adjacent to valid points, they are far from inliers with similar visual features (Fig. 3(c)). To exploit this, we extract 2D features via DINOv3 [62] and cluster them via K-Means to group pixels with similar appearance. DBSCAN is then applied within each cluster to remove spatial outliers (intra-cluster filtering), followed by refinement across clusters (inter-cluster filtering) (Fig. 3(e)). This hierarchical 2D-3D fused filtering suppresses boundary artifacts and stabilizes downstream registration.

Point Cloud Registration. We separately register the observed point clouds of the active and passive objects ($\mathcal{P}^a_{\text{obs}}$ and $\mathcal{P}^p_{\text{obs}}$) to the frames of their edited counterparts ($\mathcal{P}^a_{\text{edit}}$ and $\mathcal{P}^p_{\text{edit}}$). The pixel-aligned point clouds are defined as:

$$\begin{aligned}\mathcal{P}^{p/a}_{\text{obs}} &= \{\, \mathbf{p}^{\text{obs}}_i \in \mathbb{R}^3 \mid i \in \mathcal{M}^{p/a}_{\text{obs}} \,\},\\ \mathcal{P}^{p/a}_{\text{edit}} &= \{\, \mathbf{p}^{\text{edit}}_i \in \mathbb{R}^3 \mid i \in \mathcal{M}^{p/a}_{\text{edit}} \,\},\end{aligned} \tag{1}$$

where i indexes pixels and the superscript p/a denotes the passive or active object. A fundamental challenge lies in establishing reliable correspondences $\mathcal{C}^{p/a}$. Traditional registration [58, 76] or multi-view matching [46, 59] methods assumes geometric and appearance consistency, which breaks between current and edited states. The active object $\mathcal{O}_a$ may move, deform (e.g., "open the red drawer"), interact with $\mathcal{O}_p$, or become occluded (e.g., "insert the toast into the toaster"), leading to sparse and ambiguous matches (see details in Appendix). In contrast, image editing inherently preserves the same viewpoint and pixel-level consistency for static regions (including $\mathcal{O}_p$). Therefore for $\mathcal{O}_p$ we form dense pixel-to-pixel correspondence:

$$\mathcal{C}^p = \left\{ (\mathbf{p}^{\text{obs}}_i, \mathbf{p}^{\text{edit}}_i) \,\middle|\, i \in \mathcal{M}^p_{\text{obs}} \cap \mathcal{M}^p_{\text{edit}} \right\}, \tag{2}$$

where each observed point $\mathbf{p}^{\text{obs}}_i$ is directly paired with its corresponding edited point $\mathbf{p}^{\text{edit}}_i$ in the intersection of the two masks. For $\mathcal{O}_a$, we use semantic features f from DINOv3. Unlike geometric or low-level features, semantic features encode object-level identity and remain robust to occlusion, partial observations, and deformation. For each edited point $\mathbf{p}^{\text{edit}}_j$, we find its nearest neighbor in $\mathcal{P}^a_{\text{obs}}$ by cosine feature distance $\text{dist}(\cdot,\cdot)$, filtering out pairs whose distance exceeds a threshold $d_{\text{thr}} = 0.3$:

$$\mathcal{C}^a = \left\{ (\mathbf{p}^{\text{obs}}_{i^*}, \mathbf{p}^{\text{edit}}_j) \,\middle|\, \begin{array}{l} i^* = \arg\min\limits_{i\in\mathcal{M}^a_{\text{obs}}} \text{dist}(f^{\text{obs}}_i, f^{\text{edit}}_j), \\ j \in \mathcal{M}^a_{\text{edit}},\ \text{dist}(\mathbf{p}^{\text{obs}}_{i^*} - \mathbf{p}^{\text{edit}}_j) < d_{\text{thr}} \end{array} \right\}. \tag{3}$$

With $\mathcal{C}^a$ and $\mathcal{C}^p$ we estimate the transformation for each object using the Umeyama algorithm [73], solving for rotation $\mathbf{R}_{a/p} \in \text{SO}(3)$, translation $\mathbf{t}_{a/p} \in \mathbb{R}^3$ and scale $s_{a/p} \in \mathbb{R}_+$:

$$s_{a/p} \cdot \mathbf{R}_{a/p}\mathcal{P}^{a/p}_{\text{obs}} + \mathbf{t}_{a/p} \approx \mathcal{P}^{a/p}_{\text{edit}}. \tag{4}$$

Scale Alignment. Although the registration yields two reasonable transformations for $\mathcal{O}^a$ and $\mathcal{O}^p$, they are estimated under potentially different scales ($s_a \neq s_b$). When transformed back to the world frame, this scale inconsistency causes noticeable offsets (Fig. 4(b)), which cam impair precise manipulation. Since the passive object's pixel-level

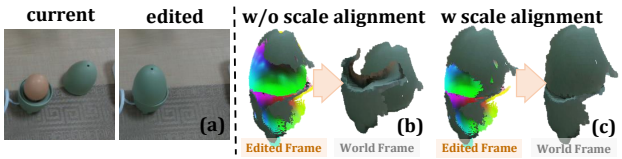

Figure 4. Illustration of scale alignment. Colorful points are $\mathcal{P}_{\text{edit}}$ with DINO features after PCA, while green points are $\mathcal{P}_{\text{obs}}$.

alignment provides a more reliable scale mapping between observed and edited frames, we treat it as the reference and enforce a unified scale by setting $s_a = s_p$. We then recompute the active object's rotation $\mathbf{R}_a$ and translation $\mathbf{t}_a$ under this shared scale (Fig. 4(c)). Notably, the original scale gap is actually small (typically <0.5), further suggesting that image editing preserves strong spatial coherence across states. To obtain the final world-frame transformation $\mathbf{T}_a$ of the active object, we first compute its scale-free relative transformation with respect to the passive object in the observation frame $[\mathbf{R}^o_{a|p} \mid \mathbf{t}^o_{a|p}]$:

$$\mathbf{R}^o_{a|p} = \mathbf{R}_p^{-1}\mathbf{R}_a, \quad \mathbf{t}^o_{a|p} = \mathbf{R}_p^{-1}(\mathbf{t}_a/s_a - \mathbf{t}_p/s_p). \tag{5}$$

We transform this relative motion into the world frame using the observation-to-world transformation $[\mathbf{R}_{\text{o2w}} \mid \mathbf{t}_{\text{o2w}}]$:

$$\begin{aligned} \mathbf{R}^{\text{w}}_{a|p} &= \mathbf{R}_{\text{o2w}}\mathbf{R}^o_{a|p}\mathbf{R}^T_{\text{o2w}}, \\ \mathbf{t}^{\text{w}}_{a|p} &= \mathbf{R}_{\text{o2w}}\mathbf{t}^o_{a|p} + \mathbf{t}_{\text{o2w}} - \mathbf{R}^{\text{w}}_{a|p}\mathbf{t}_{\text{o2w}}. \end{aligned} \tag{6}$$

Finally, the active object's transformation in the world frame is given by $\mathbf{T}_a = [\mathbf{R}^{\text{w}}_{a|p} \mid \mathbf{t}^{\text{w}}_{a|p}]$.

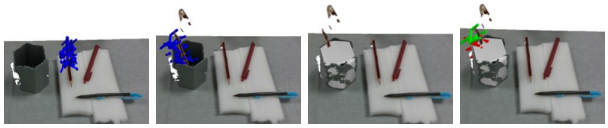

Figure 5. Illustration of edited-informed grasping.

3.4. Edited Goal Conditioned Execution

We split manipulation into two stages, approach and transform, to convert the transformation into motion primitives.

Edited-informed Grasping. Off-the-shelf grasping method, AnyGrasp [19], generates numerous grasp candidates for a target object. However, blindly adopting these grasps is often unsuitable in task-specific manipulation. For example, "insert the pen into the holder" requires grasping the pen from its tip, not its body, to avoid future collision with the holder. To address this, we leverage the edited future state and compute the convex hull of the passive object's point cloud (Fig. 5(c)). Then we transform each grasp candidate via the estimated transformation $\mathbf{T}_a$ (assuming the gripper and active object remain rigidly attached pose-grasp) (Fig. 5(a)(b)). We discard any grasp that would collide with the passive object's convex hull in the edited state, thus retaining only task-feasible grasps (Fig. 5(d)).

Primitive-based Motion Planning. While one could leverage LLMs/VLMs to generate task-specific waypoints, our focus lies on spatial-prior extraction. Therefore, we use a heuristic motion-primitive approach to convert the target transformation into executable robot motion. Specifically, a single intermediate waypoint is defined based on task type and target transformation, followed by motion planning using Curobo [68]. Motion primitives are parameterized action units (e.g., "lift by dz") and these have been widely used in manipulation [33]. For tasks such as insertion, placing, stacking, or covering, we lift the target transformation by a small offset dz as the waypoint; for pouring, we set the waypoint by combining the current rotation with the target translation. This compact formulation covers a wide variety of single-step manipulation tasks without relying on large-scale demonstration or video-based imitation.

4. Experiment

In this section, we evaluate and analyze LAMP to address three key questions: **(1)** How well does our image-editing-based zero-shot registration perform in aligning manipulation pairs (Sec. 4.1)? **(2)** To what extent can our editing-based manipulation framework generalize in open-world scenarios (Sec. 4.2? **(3)** Can the image-editing prior support robust and long-horizon manipulation (Sec. 4.3)?

4.1. Point Cloud Registration for Manipulation

Tasks. To evaluate our registration method on manipulation pairs, we collect real-world scenes captured using a single-view RGB-D camera, as no existing one meets our needs. Each scene contains an active object $\mathcal{O}_a$, a passive object $\mathcal{O}_p$, and a natural language instruction describing the interaction. Given the two partial point clouds and the instruction, the task is to predict the relative 6-DoF transformation of $\mathcal{O}_a$ with respect to $\mathcal{O}_p$ that fulfill the described manipulation. Collected pairs covers diverse manipulation types (e.g., insertion, covering, placing, assembling, cutting).

Baselines. We compare our method with two point cloud-based methods: 1) **Two by Two (2BY2)** [54], which predicts relative transformations between two object point clouds via a two-step SE(3) pose-estimation pipeline for multi-task assembly, 2) **AnyPlace** [86], which predict placement poses from local point clouds cropped at VLM-proposed locations.

Results. As shown in Fig. 6, LAMP generalizes well across diverse manipulation tasks and is markedly more robust to noisy, partial point clouds than all baselines. This advantage mainly stems from the strong spatial priors embedded in image-editing models. Our proposed reasoning mechanism further lifts these implicit 2D constraints into coherent 3D relationships, enabling reliable alignment under real-world noise and occlusions. In contrast, both AnyPlace [86] and 2BY2 [54] struggle to generalize across tasks. AnyPlace is fine-tuned on point clouds from simulation environments for tasks such as insertion, stacking, hanging, and plac-

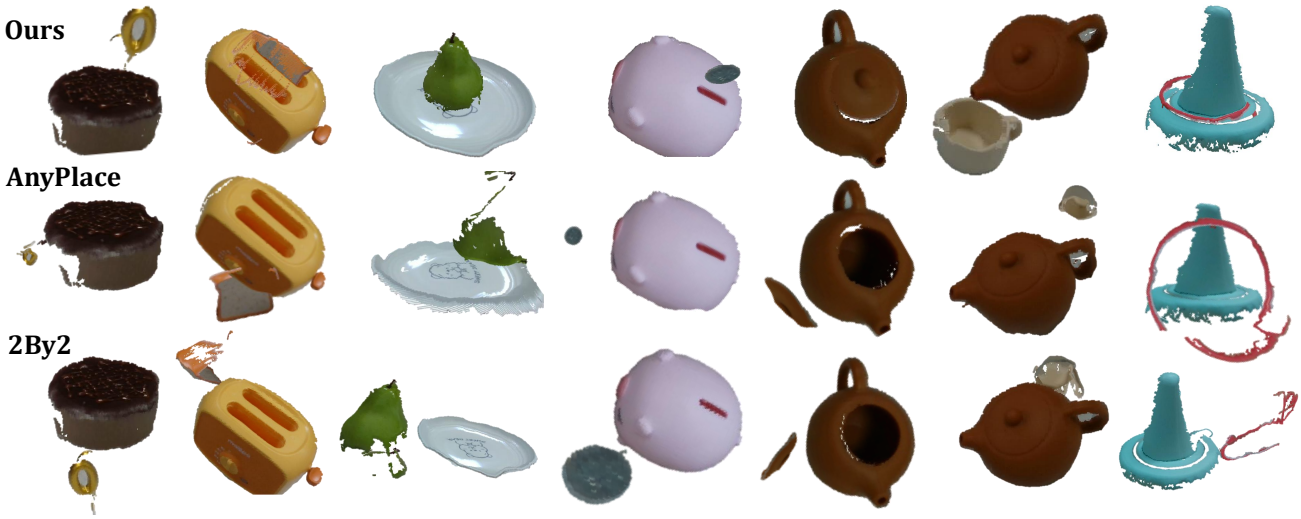

Figure 6. **Qualitative results of point cloud registration across diverse manipulation tasks.** LAMP consistently aligns active and passive objects under various task configurations, showcasing strong generalization and robustness to noisy, partial real-world point clouds.

Table 1. Success rate of 13 real-world manipulation tasks. '/' indicates the method is not applicable for that task.

Tasks	VoxPoser	CoPa	Rekep	Ours
Egg placing	2/10	2/10	4/10	**6/10**
Coin insertion	0/10	0/10	0/10	**5/10**
Pencil insertion	0/10	4/10	3/10	**7/10**
Toast insertion	0/10	0/10	0/10	**6/10**
Lid covering	0/10	3/10	4/10	**8/10**
Pen-cap covering	0/10	1/10	2/10	**6/10**
Tea pouring	0/10	1/10	3/10	**6/10**
Toast cutting	0/10	0/10	5/10	**8/10**
Block assembly	0/10	1/10	0/10	**6/10**
Ring stacking	2/10	1/10	3/10	**8/10**
Total	4.0%	13.0%	24.0%	**66.0%**
Drawer opening	2/10	4/10	/	**6/10**
Drawer closing	4/10	4/10	/	**7/10**
Toaster opening	2/10	1/10	/	**5/10**
Total	26.7%	30.0%	/	**60.0%**

ing, while 2BY2 is trained on mesh-sampled point clouds for insertion, covering, and placing. Their dependence on clean, task-specific training data limits their transferability to noisy and incomplete real-world observations, leading to a noticeable generalization gap. These results highlight that image-editing priors offer a strong and transferable spatial understanding that enables robust point cloud registration across unseen tasks and real-world variations.

4.2. Open-world Manipulation

Hardware Configuration. Our experiments are conducted on a UFACTORY xArm7 robotic arm equipped with its UFACTORY xArm Gripper G2. An Intel RealSense D435i RGB-D camera is mounted opposite the robot to capture a third-person view of the workspace.

Tasks and Metrics. We evaluate the open-world manipulation capability of LAMP across a diverse set of everyday object-centric tasks, covering aspects from high-precision manipulation to articulated-object manipulation. In total, we select 13 representative tasks, including egg placing, coin insertion, pencil insertion, toast insertion, lid covering, pen-cap covering, tea pouring, toast cutting, block assembly, ring stacking, drawer opening and closing, and toaster opening. Each task is executed for 10 trials with random object poses, and overall success rates are reported in Tab. 1 with more details in the appendix.

Baselines. As analyzed in Sec. 4.1, **Two by Two** [54] and **AnyPlace** [86] generalize poorly to single-camera, real-world manipulation setups. Therefore, we compare our method with three additional zero-shot open-world manipulation baselines: 1) **Voxposer** [30], which uses LLM-generated code to build 3D value maps conditioned on language instructions for trajectory synthesis; 2) **CoPA** [29], which employs VLMs to infer spatial constraints between interaction keypoints and interaction surface vectors; and 3) **Rekep** [31], which formulates VLM-predicted relational keypoints as cost terms for trajectory optimization. We always provide CoPA with best available masks.

Results. LAMP exhibits strong performance in task diversity, fine-grained manipulation, and execution robustness compared with baselines. This advantage stems from implicit 2D spatial cues in edited images, which are effectively lifted into 3D transformations through our object-centric formulation. Qualitative results in Fig. 7 and Fig. 8 illustrate these strengths. We analyze the performance from two perspectives: **the limits of language-based constraints** and **the challenges of input representations**. Language-based constraints suffer from sparse and ambiguous 3D guidance, missing fine-grained relations (i.e., rotations, contact geometry, and object-to-object alignment) and thus leading to failures in precision tasks such as toast or coin insertion (Fig. 8). For geometry-sensitive tasks like egg placing or

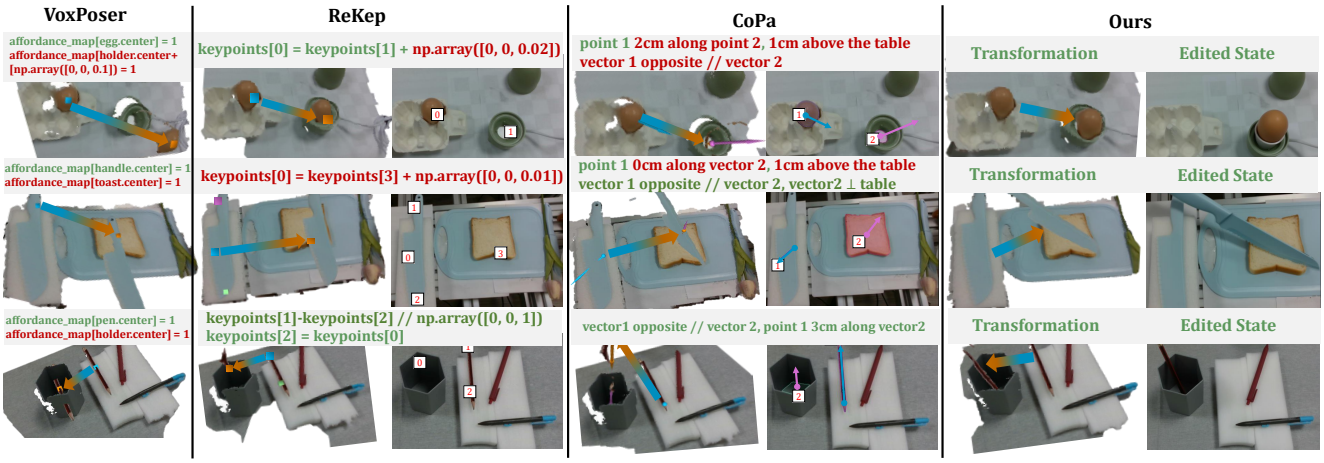

Figure 7. **Qualitative comparison of different manipulation representations.** The blue to orange arrows indicate the target manipulation pose. Voxposer [30] grounds manipulation at the center of the object, ReKep [31] uses keypoints, CoPa [29] uses keypoints and vectors, and our approaches uses a full 3D inter-object transformation.

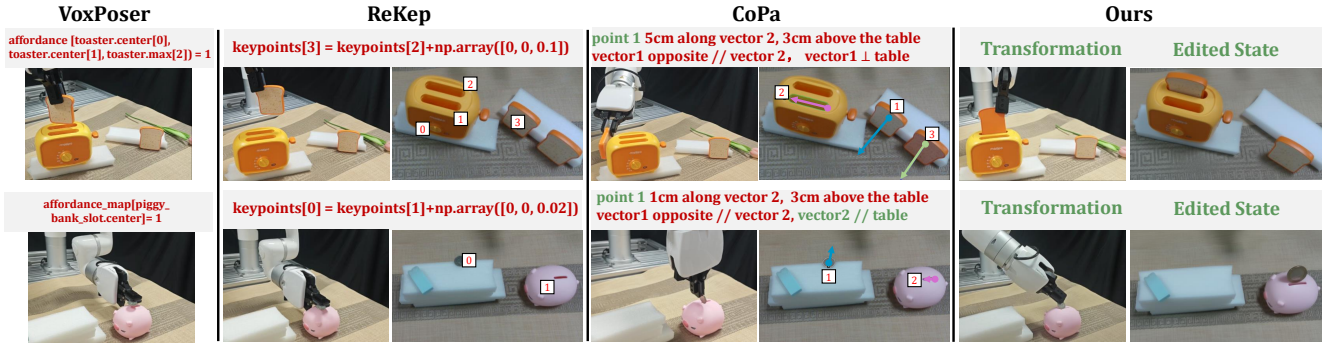

Figure 8. **Qualitative results on real-world insertion tasks** (*Toast, Coin*). Voxposer [30] fails to infer rotations; ReKep [31] misidentifies keypoints and rotations; CoPa [29] cannot reliably capture vector constraints; our method recovers precise inter-object 3D transformations.

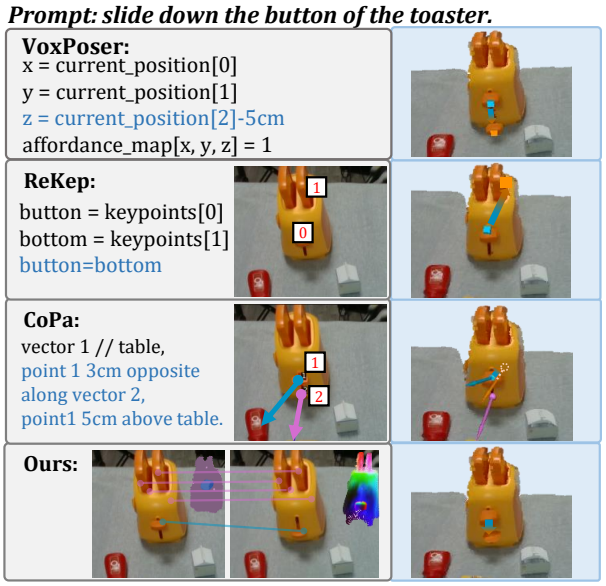

Figure 9. **Qualitative analysis of articulated manipulation.**

knife cutting (1st and 2nd row of Fig. 7), VoxPoser [30] and ReKep [31] exhibit limited rotational awareness, while CoPa may produce contradictory constraints due to weak geometric understanding. In contrast, our method uses edited images to provide implicit spatial priors that encode both the rotation and interaction regions of objects, enabling accurate 3D alignment even for fine-grained manipulation. Beyond language limitations, the input modality itself also constraints performance. VoxPoser can convert phrases like "slide down" into z-axis motion (1st row of Fig. 9) but cannot infer metric geometry without visual grounding (e.g., -5cm); ReKep may misidentify keypoints without task-specific keypoint extraction (e.g., misidentified "bottom" keypoint in 2nd row of Fig. 9). CoPa projects 3D vectors onto 2D observation, making it sensitive to noisy point clouds (e.g., incorrect surface normal of the button in 3rd row of Fig. 9) and ambiguous shapes (i.e., ellipsoids like eggs in 1st row of Fig. 7). Our method leverages subject consistency and visual correspondence between current and the edited states (4th row in Fig. 9), providing a robust global context that generalizes across diverse object geometries and articulated-obejct tasks.

4.3. Long-horizon Manipulation

Following the setup in Sec. 4.2, we further evaluate LAMP on long-horizon manipulation tasks to demonstrate its un-

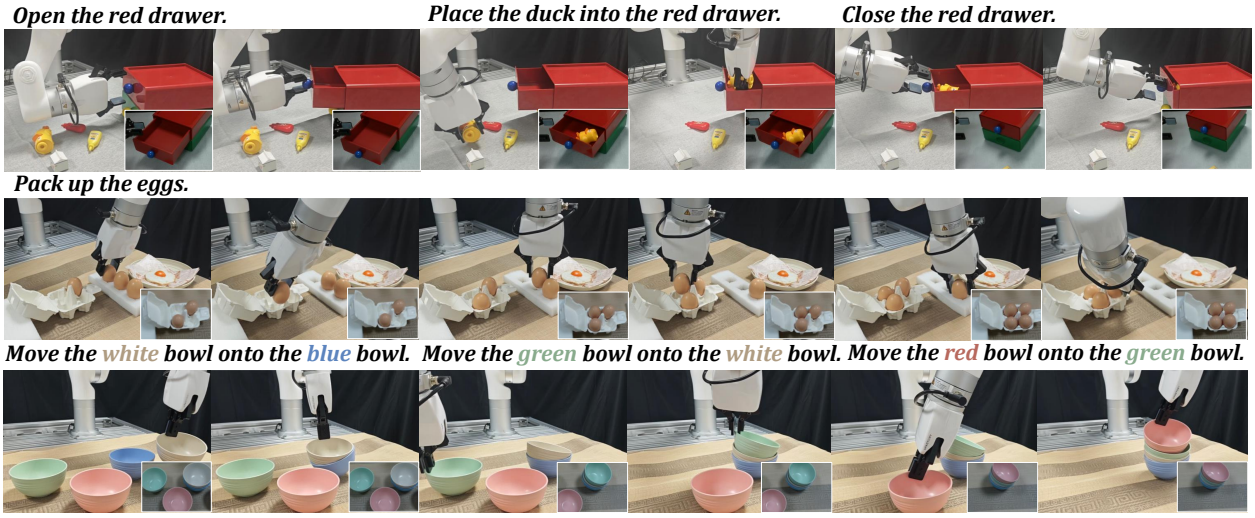

Figure 10. **Example rollouts of long-horizon manipulation tasks.** Bottom right corner shows the edited prior for each step.

Table 2. Quantative results of viewpoint influence (*ring stacking*).

View-point	ReKep	CoPa	Ours (wo filter)	Ours (wo scale)	Ours (full)
0°	1/10	1/10	2/10	3/10	**6/10**
45°	3/10	4/10	6/10	3/10	**8/10**
90°	2/10	6/10	7/10	4/10	**8/10**

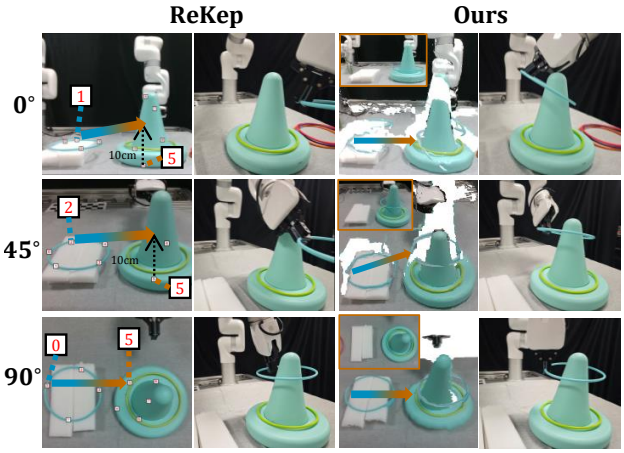

Figure 11. Qualitative analysis of camera-viewpoint effects.

derstanding of multi-step object-centric interactions. Long-horizon tasks typically require decomposition into subtasks, where the execution of each step depends on the final state of the previous one. We design three long-horizon tasks: putting a duck into the red drawer, packing the eggs, and setting up the table. Fig. 10 shows example execution rollouts, highlighting that LAMP maintains accurate object alignment and successfully completes each subtask in sequence. To further analyze the benefits of our approach, we compare the use of spatial priors from edited images against video generation priors. Edited-image priors exhibit stronger adherence to semantic constraints and better background consistency, resulting in more reliable and coherent long-horizon manipulation (see details in appendix).

4.4. Ablation Study

We ablate our pipeline on the ring stacking task under three viewpoints (0°, 45°, and 90°) (more in appendix). Success rates over 10 trials are in Tab. 2 and qualitative results are shown in Fig. 11. Without our proposed point cloud filtering, the success rate at the 0° viewpoint drops a lot, as the ring becomes almost line-like in the image, making depth highly unreliable (1st row in Fig. 11. Removing scale alignment also degrades performance, since stacking requires precise relative placement and inconsistent scales break this alignment. Compared with baselines, our approach is notably more robust to viewpoint changes, as illustrated in Fig. 11. ReKep [31] and CoPa [29] both rely on relationships between 2D keypoints or projected vectors, which are inherently limited by the field of view and depth accuracy of corresponding keypoints. In contrast, our method lifts the implicit spatial priors from edited images into full 3D transformations and performs dense registration, leading to greater resilience to noise, occlusion, and partial geometry.

5. Conclusion

In this work, we present LAMP, a generalizable representation that lifts image editing as 3D priors to extract inter-object transformations. Leveraging implicit spatial cues in edited images, LAMP provides precise 3D relational understanding, enabling robust generalization across viewpoints, object geometries, and fine-grained manipulation tasks. This work marks a promising step toward scalable open-world manipulation. Despite these promises, limitations remain. LAMP currently handles rigid-body interactions and does not address soft-body or deformable-object manipulation. The framework relies on motion planning to execute and thus tasks requiring intermediate trajectories may need additional motion priors or task-specific planning heuristics. As with most language-based models, it requires moderate prompt engineering to ensure consistent edits.

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